ORBCOMM®

WHERE DATA DRIVES DECISIONS



ST 21xx
Hardware Guide

T412, Version 09
Jan 2025

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ABOUT THIS GUIDE

Purpose

This document provides an overview of the hardware characteristics and specifications for the ST 2100 and its variants, referred to as "ST 21xx".

Notation

Hardware components and hardware stickers in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Numbered lists indicate a series of steps required to complete a task or function.

Bulleted lists highlight information where order or sequence is not crucial.

Reference

The content in the following Technical Publications may be useful in conjunction with this guide. These documents are available from the partner-support portal, the ORBCOMM Developer Toolkit, or Support.

The names, numbers, and content of these documents are subject to change without notice. Always check the appropriate category on the Partner Support Portal for the most current version of each document.

[T411]	ST 21xx Installation Guide
[R403]	ST 21xx Development Kit User Guide
[N215]	OGx Network Overview
[N200]	IsatData Pro Network Services Overview
[T415]	AT Interface Developer Guide - OGx
[T403]	AT Interface Developer Guide - IsatData Pro
[T405]	LSF API Reference Guide

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1 OVERVIEW

Each ST 21xx consists of a single environmentally sealed mechanical enclosure containing an integral antenna, a satellite modem for communicating with the satellite, its own power supply circuitry, an integral GPS subsystem, three digital output and one digital input port, and an RS-232 port. A supercapacitor provides last gasp capability and brown out protection.

Figure 1: ST 21xx



1.1 ST 21xx

The ST 21xx (Table 1) operates on the satellite network and is self-contained, environmentally sealed, compact, and provides low power consumption.

Table 1: ST 21xx Models

Model	Pins	Configuration
ST 2100	8-pin	side or bottom mount with supercapacitor

The ST 21xx offers two configurations: bottom mount or side mount (see figures below).

Figure 2: ST 21xx as a Side Mount (Surface Mount)



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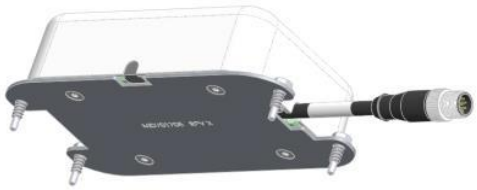
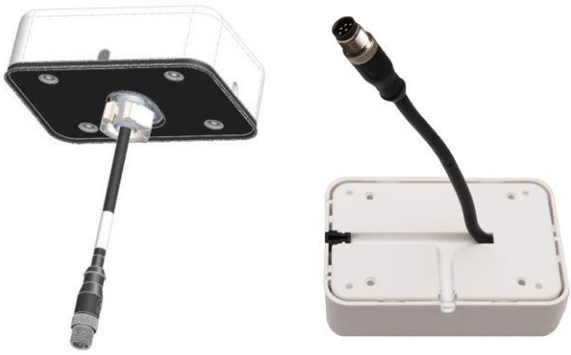


Figure 3: ST 21xx as a Bottom Mount (Cab Mount)



1.2 Key Features and Benefits

The ST 21xx has the following key features and benefits:

- Designed to be incorporated into a Solution Provider's (SP) solution
- Built-in multi-GNSS receiver to calculate position, speed and heading with SP-preferred constellation
- Quick and easy installation reduces labor time and costs
- Very low power consumption enables a wide variety of autonomous applications
- Ships with installed firmware that is over-the-air upgradeable for significant cost and schedule savings, and convenience
- Wide operational temperature range to support a very broad range of applications and geographies
- Rugged construction to support applications in the harshest environments
- Two-way satellite service
- Up to 6,399 bytes from-mobile messages
- Up to 10,000 bytes to-mobile messages

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1.3 Operational States

The device has several operational states.

Table 2: Operational States

Operational States	Description
Transmit	In transmit state the device is transmitting a signal to the satellite. If the device has a message to send, it automatically switches to Transmit state.
Receive	In receive state the device is attempting or actively listen to the satellite. The time interval between receives is configurable to manage power consumption. Receive state is mutually exclusive with transmit state. The network ensures that transmit and receive signals do not collide.
Idle	If the device is not transmitting or receiving and an external host is connected to its serial port, it is in the idle state. In this state, the device is still able to receive, process, and answer AT commands as required.
Sleep	If the device is not transmitting or receiving and no external host is connected to its serial port, it is in the sleep state. In this state, the device draws minimal current. It monitors its external digital input and serial port for activity and reacts as required by transitioning to other states with no loss of information.

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2 COMPLIANCE

The ST 2100 has received the following certifications:

Inmarsat Type Approval

FCC Part 25

CAUTION: Mount the ST 2100 at least 20 cm (8 in.) away from humans.

CE Mark RED

Ingress Protection

- IP67

RoHS

- Restriction of Hazardous Substances (RoHS) [1]

[1] European Union's (EU) Directive 2002/95/EEC "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment.

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3 SPECIFICATIONS

3.1 Temperature

Parameter	Value
Operating Temperature	-40° to +85°C (-40° to +185°F)
Storage Temperature	-40° to +85°C (-40° to +185°F)

3.2 Temperature Sensor

The temperature sensor monitors the ambient temperature of the device.

The sensor can measure an internal temperature over the range of -40 to +85°C (-40° to +185°F), to an accuracy of ±0.3°C (±0.6°F).

3.3 Connector

The ST 21xx uses a standard circular M12 connector, A-coded, 8-pin, male (IEC 61076-2-101).

3.3.1 Connector Pin Assignment

Figure 4: ST 21xx Connector Pin Assignment (plug contacts)

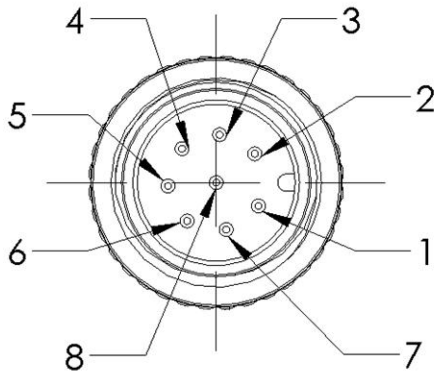


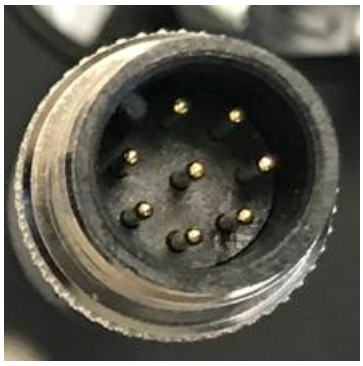
Table 3: Electrical Pin Assignment (ST 21xx)

Pin	Functionality
1	External Reset Input
2	Ground
3	RS-232 Rx (Input)
4	RS-232 Tx (Output)

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Pin	Functionality
5	VEXT
6	1 PPS Output
7	Event Notification Output
8	External Reset Output

Figure 5: View of ST 21xx Male Connector



3.4 Power

3.4.1 Input Range

Parameter	Value
Power Supply Voltage	9 to 32 V DC
Reverse Polarity Protection	-40 V maximum

3.4.2 Power Consumption

Table 4: Power Consumption

Parameter	-40°C/-40°F		25°C/77°F		85°C/185°F		Units
	12 V	24 V	12 V	24 V	12 V	24 V	
Low Power Sleep	118.6	117.7	120.7	114.8	152.6	131.6	µA
Idle (charger off [1])	14.6	7.7	13.4	7.5	13.9	7.9	mA
Idle (charger on[1])	207	111.3	202	109	199.4	108.8	mA
GNSS on	35.8	19.9	34.9	19.4	37	20.5	mA
Receiver on	75.3	41	76.2	40.8	77.9	42.6	mA
Transmitter on	552.8	272.1	571.45	279.9	609.2	297.9	mA

[1]Only applies to an ST 2100 with a supercapacitor.

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3.4.3 Cable Length Recommendations

Power

To maximize cable length, low resistance cabling must be used, 25.67 Ohm / 1000 ft. or better, to minimize voltage drop across the cable during high current draw. The following table shows the maximum cable length versus input voltage using a 25.67 Ohm / 1000 ft. cable.

Power					
Supply Voltage (V)	**Cable Wire Gauge (AWG)**	**Cable Resistance (Ohm / ft)**	**Max. Current Drawn (A)**	**Cable Length (ft.)**	**Cable Length (m)**
9	24	0.026	0.882	46	14.02
12	24	0.026	0.609	197	60
24	24	0.026	0.298	1363	415
32	24	0.026	0.232	2726	831

* Length of cable between the external power source and the ST 2100.

RS-232

The maximum recommended cable length for RS-232 operation is 50 ft. (15 m) using a baud rate of 9600 bps if the total capacitance of the cable is less than 2500 pF.

For example, if a CAT-5 cable is used with a capacitance of 17 pF/ft, then the maximum cable length would be 147 ft. (45 m).

For a standard shielded cable, the Data Rate and corresponding Maximum cable lengths are as follows, according to the specifications:

- 2400 bps – 200 ft. (60 m)
- 4800 bps – 100 ft. (30 m)
- 9600 bps – 50 ft. (15 m)
- 19200 bps – 25 ft. (7.6 m)
- 38400 bps – 12 ft. (3.7 m)
- 57600 bps – 8 ft. (2.6 m)
- 115200 bps – 4 ft. (1.3 m)

3.4.4 Surge Current

The surge current refers to the maximum instantaneous current drawn from external power when external power is initially applied to the ST 21xx.

	-40°C/-40°F		**25°C/77°F**		**85°C/185°F**			
Parameter	**12 V**	**24 V**	**12 V**	**24 V**	**12 V**	**24 V**	**Units**	**Pulse Duration**
Surge Current	3.24	7.56	5.4	7.32	3.24	6.84	A	<50 µs

3.4.5 Load Dump Protection

The ST 21xx's power supply input voltage range is 9 to 32 V. Above this range the load dump protection circuitry disconnects external power so that the ST 21xx is not damaged. The cut-off threshold is ±34 V. External power is

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disconnected during the load dump pulse.

3.4.6 Fuse/Over Current Protection

The device has an internal 5 A fast acting fuse, which provides protection in the event of an internal short on the device.

3.4.7 Inrush Current

Power On events occur on modem power on and restart for satellite beam searches and ahead of receive and transmit events.

Table 5: Inrush Current on Input Power @12V

Operation Mode	Inrush Current on Input Power		
	Amplitude (mA)	Period (ms)	Charge (mC)
Power On	2580	0.508	1.3
Receive On	208	0.116	0.069
Transmit On	920	0.167	1.18

3.4.8 Supercapacitor

This section only applies to an ST 2100 with a supercapacitor.

A supercapacitor with a capacity of 5F provides last gasp capability and brown out protection. When external power drops due to a brown out condition or if the external power is turned off the supercapacitor allows the device to continue operating or in the case of a power loss continue operating for a sufficient period to send off a last gasp message.

The ST 2100 can measure the supercapacitor voltage over the range of 1.71-4.1V, to an accuracy of ± [5% + 150 mV].

Once the supercapacitor is depleted, it takes about 25 seconds to charge.

3.4.8.1 Supercapacitor Charger

Table 6: Supercapacitor Charger Specifications

Parameter	Condition	Min. Value	Typ. Value	Max. Value	Units
Supercapacitor operating voltage range	Backup mode	1.8	-	4.2	V
System operating voltage range	Backup mode	-	-	4.4	V
Supercapacitor charge voltage	High temperature de-rating applied	-	-	4.2	V
Under voltage lockout threshold	-	1.6	-	1.7	V
Supercapacitor charging current	Limit set in hardware	-	-	400	mA
Charger input leakage	Affect on supercapacitor	-	-	50	nA
Charger quiescent current	Burst mode, end of charge	-	-	40	µA
Peak current limit in backup operation	-	5	6	7	A
DC current limit in backup operation	-	3.5	4.5	-	A

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Parameter	Condition	Min. Value	Typ. Value	Max. Value	Units
Soft start-up time in backup mode	-	0.8	1.3	1.8	ms
Oscillator frequency	-	0.9	1.2	1.5	MHz

3.5 Input/Output

The ST 21xx has one digital input and three digital output ports:

- Reset Input
  - The Reset Input allows an external controller to reset the ST 21xx.A high level applied to this input for at least 400 ms will result in the ST 21xx processor resetting. Reset is an active high CMOS/TTL input.
- 1 PPS Output
  - The 1 PPS Output is available from the GNSS module. It outputs a 100 ms pulse every second, providing a valid fix is present. If the GNSS signal is blocked, the 1 PPS stops.
  - By default, the GNSS is only on when requested by the application or the network. For constant time updates, the GNSS must always be on (GNSS Continuous Mode). 1 PPS is an active high CMOS/TTL output. Refer to [T403] for further details.
- External Reset Output
  - The External Reset Output allows the ST 21xx to reset an external circuit. It outputs a 250 ms pulse upon receiving a reset (SIN 0, MIN 68) to-mobile message with fields set accordingly. This allows users to send a to-mobile message to reset their external controller remotely. External Reset is an active high CMOS/TTL output. Refer to [T403] for further details.
- Event Notification Output
  - The Event Notification Output indicates that one or more configured events have occurred. Example events include incoming satellite messages, modem reset, new GPS position, and transmit compete. It is possible to specify which events cause this output to become active by writing to S-register 88. This output is set to high when one or more configured events occur and is only reset when S-register 89 is read. Event Notification is an active high CMOS/TTL output. Refer to [T403] for further details.

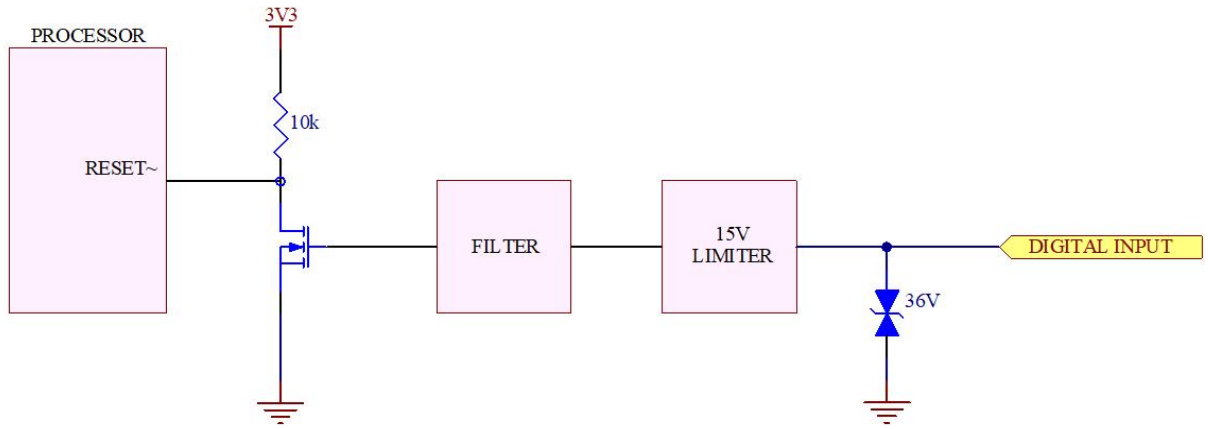
The inputs and outputs are protected against transients with a 36 V TVS.

3.5.1 Reset Input

Figure 6 shows a schematic for the I/O when configured as a digital input.

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Figure 6: Reset Digital Input



The following table describes the input specifications.

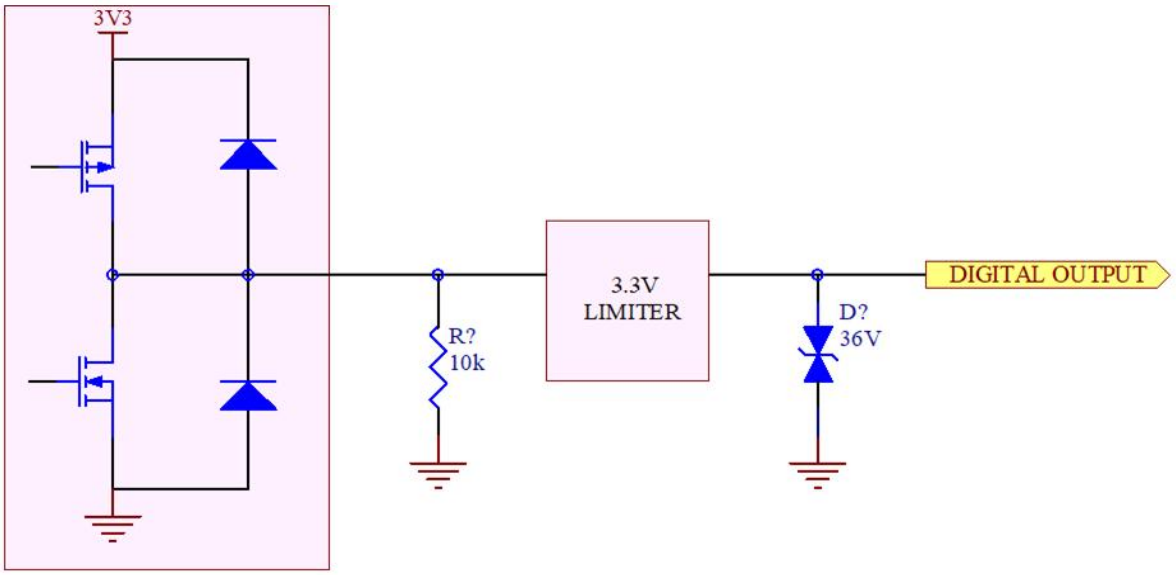
Parameter	Min.	Max.	Units
Input low range	-10	0.85	V
Input high range	1.5	32	V
Input current, Vin = 3.3 V	-	5	µA
Input bandwidth	1	1000	Hz
Minimum active high pulse width	0.4	-	s

3.5.2 Digital Outputs

Figure 7 shows the digital outputs for the ST 21xx.

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Figure 7: Digital Output



The following table describes the output specifications.

Parameter	Min.	Typ.	Max.	Units
Output high voltage - open circuit	3.14	3.3	3.46	V
Output high voltage (sourcing 25 µA)	3.24	-	-	V
Output low voltage (sinking 25 µA)	-	-	0.05	V
Voltage limits (survivability)	-10	-	32	V
Output current	-	-	0.45	mA

3.6 Serial Interface

3.6.1 RS-232

The half-duplex RS-232 port has the following settings: 9600 bit/s (8 data, no parity, 1 stop bit). The baud rate is configurable up to 115,200 bps.

The electrical characteristics of the interface are:

Parameter	Min.	Typical	Max.	Units
Serial Rx Input High Voltage Threshold	-	1.5	2.4	V
Serial Rx Input Low Voltage Threshold	0.6	1.2	-	V
Serial Tx Output Voltage Swing	±5.0	±5.4	-	V
Serial Tx Output Short Circuit Current	-	±35	±60	mA
Serial Tx Output Leakage Current	-	-	±25	µA

3.6.1.1 Auto RS-232 Power On/Off

The device uses the input voltage to turn the RS-232 driver on/off.

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Parameter	Value
Valid Rx Input Threshold	Rx > 2.7 V or Rx < -2.7 V
Invalid Rx	-0.3 V < Rx < 0.3 V

3.7 RF Specifications

3.7.1 Frequency

Parameter	OGx Value	IsatData Pro Value
Receive		
Frequency Band	1525 to 1559 MHz	1525 to 1559 MHz
Modulation	OQPSK	OQPSK
Symbol Rate	2000, 8000, 16000 symbols/seconds	3000 symbols/second
Polarization	RHCP	RHCP
Transmit		
Frequency Band	1626.5 to 1660.5 MHz	1626.5 to 1660.5 MHz
Modulation	OQPSK	OQPSK
Symbol Rate	800, 1600, 3200, 6400 12800 symbols/seconds	900 symbols/second (maximum)
Polarization	RHCP	RHCP

3.7.2 Antenna

Parameter	Value
Maximum EIRP	7 dBW
Elevation Angle	20 degrees
Maximum Transmit Antenna Gain	5 dBic

3.8 GNSS

The manufacturer's specifications are shown in the table below.

Table 7: Multi-GNSS Specifications

Parameter	GPS	GLONASS	BeiDou	GPS/GLONASS
Time to First Fix[1]				
Cold Start	29 s	30 s	34 s	26 s
Hot Start	1 s	1 s	1 s	1 s
Sensitivity				
Tracking	-166 dBm	-166 dBm	-160 dBm	-167 dBm

[1] All satellites at -130 dBm

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Parameter	GPS	GLONASS	BeiDou	GPS/GLONASS
Reacquisition	-160 dBm	-156 dBm	-157 dBm	-160 dBm
Hot Start	-157 dBm	-156 dBm	-155 dBm	-157 dBm
Cold Start	-148 dBm	-145 dBm	-143 dBm	-148 dBm
Horizontal Position (CEP) [1]	2.5 m	4.0 m	3.0 m	2.5 m
Accuracy				
Velocity	0.05 m/s			
Heading	0.3 degrees			

3.9 Mechanical

The ST 2100 mechanical enclosure is a rugged, impact, and chemical resistant plastic material.

Figure 8: ST 2100 Top View - Side Connector Dimensions

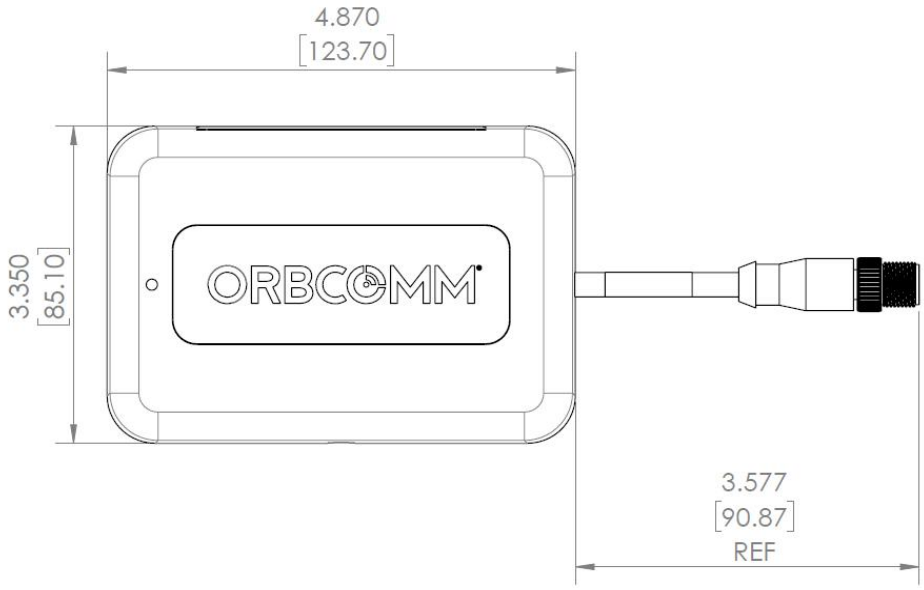
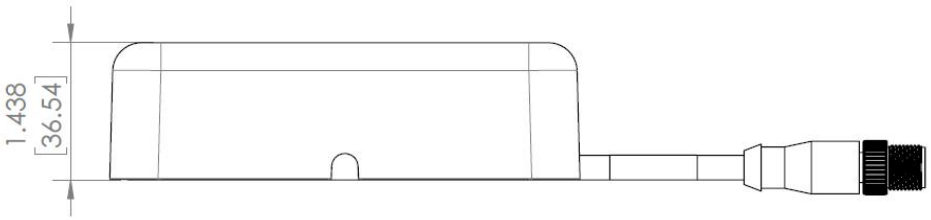


Figure 9: ST 2100 Side View Dimensions



[1] CEP, 50%, 24 hours static, -130 dBm

Figure 10: ST 2100 Bottom Connector Cable Length

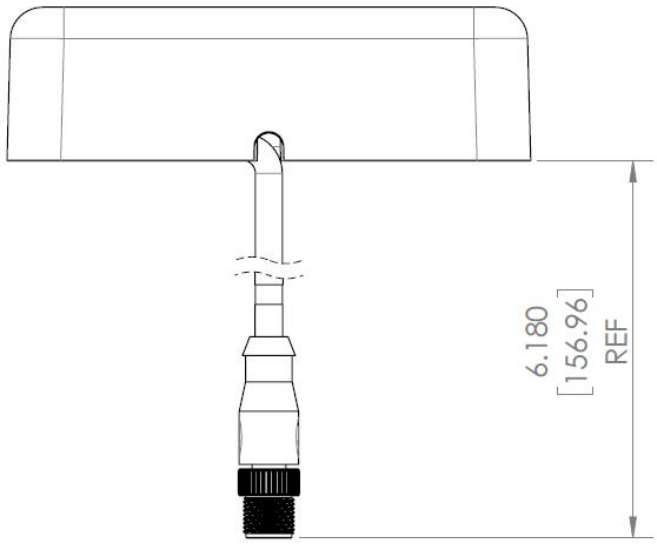


Table 8: Mass and Materials

Parameter	Value
ST 2100 mass	245 g (0.54 lbs)
Enclosure Material	Lexan EXL9330 Resin

3.10 LED

The ST 21xx has an integral LED to indicate that the ST 21xx has successfully powered on, otherwise it is off.

Parameter	Value
Color	Red

3.11 Nonvolatile Storage

The device has nonvolatile flash memory, which is shared by both the device firmware and user services.

Parameter	Value
Nonvolatile Onboard Flash Storage	8 MB
Write-Erase Cycles (per operating life)	100 000

3.12 RAM

Parameter	Value
PSRAM (volatile memory)	4 MB

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3.13 Environmental

Parameter	Description
Vibration	The ST 2100 meets all its specifications during exposure to random vehicular vibration levels per SAE J1455, section 4.10.4.2 and MIL-STD-810G, section 514.6, fig 514.6C-1.
Mechanical Shock	The ST 2100 meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 20G and durations of 11 ms as specified in MIL-STD-810G, section 516.6, Procedure I, section 2.3.2c, 3/axis/(positive and negative direction).
Altitude	The ST 2100 meets all its specifications after a nonoperating 12.2 km altitude test as detailed in SAE J1455, section 4.9.3, except with an ambient temperature of -40°C.
Thermal Shock	The ST 2100 meets all its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2
Drop Test	The ST 2100 meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1.
ESD	The ST 2100 meets all its specifications after exposure of the enclosure to 2 kV ESD contact discharge per IEC 60945 and IEC61000--4-2, level 3. All the connections on the 8-pin harness connection except for the VEXT power rail are ESD protected to ±30 kV contact discharge per IEC 61000-4-2 far exceeding level 4.

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APPENDIX A ST 21XX BLUNT CUT CABLE

Details for the ST 21xx blunt cut cable (ST101349-001) are provided below.

- 24 AWG 8C al braid jacket black PVC cable, overall diameter 6.5 mm (0.25")
- M12 A coding 8-pin female connector with shield
- 5 m (16') length

Figure 11: ST 21xx Blunt Cut Cable (J2)

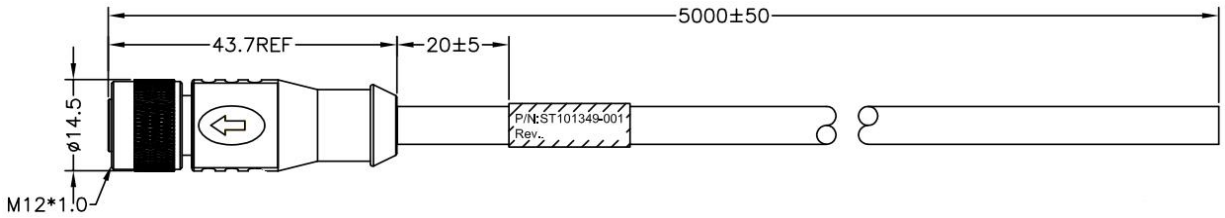


Figure 12: Blunt Cut Cable Connector (J1)

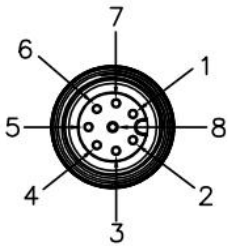


Table 9: ST 21xx Blunt Cut Cable PIN Descriptions

J1	Color	Description	J2
1	White	Ext Reset In (IN)	OPEN
2	Brown + Braid	Ground	
Shell			
3	Green	Console RS-232 Rx (IN)	
4	Yellow	Console RS-232 Tx (OUT)	
5	Gray	Vext	
6	Pink	1-PPS Out (OUT)	
7	Blue	Event Notification Out (OUT)	
8	Red	Ext Reset Out (OUT)	

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APPENDIX B ST 21XX EXTENSION CABLE

Details for the ST 21xx extension cable (840-00193) are provided below.

- Black cable jacket and 24 AWG conductors
- M12-8 Socket (Female) circular connector and an M12-8 Pin (Male) circular connector
- IP65 rated

Figure 13: ST 21xx Extension Cable

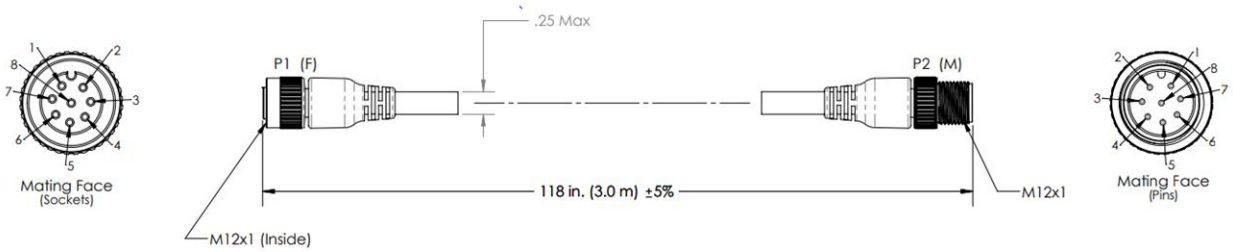


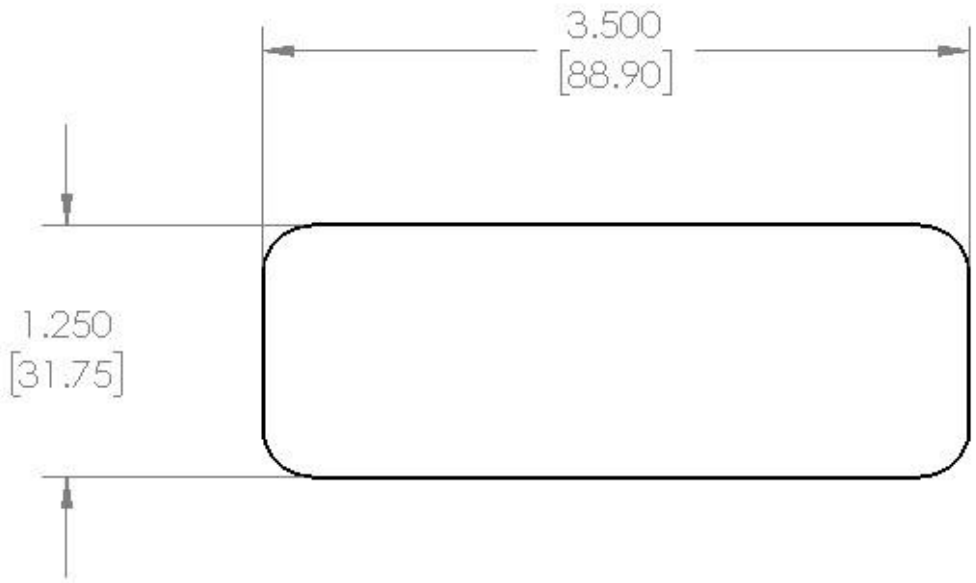
Table 10: ST 21xx Extension Cable PIN Descriptions

Wire	P1	P2	Function
1	1	1	Chassis Ground
2	2	2	Signal Ground
3	3	3	RS-232 (Txd/Rxd)
4	4	4	RS-232 (Rxd/Txd)
5	5	5	Power
6	6	6	1-PPS Out (OUT)
7	7	7	Event Notification Out (OUT)
8	8	8	Ext Reset Out (OUT)

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APPENDIX C BRANDING LABELS

Image and dimensions shown are not to scale. Please refer to the Partner Support Portal (ST Devices - mechanical files folder) to download a .DXF file.



Parameter	Value
Overlay Material	GE Lexan 8A35 Film, .010" thick, velvet finish
Adhesive Backing	3M 7952, .002" thick

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